

2019



Group B [6x5=30]



Attempt any SIX questions.

2. Differentiate Object Oriented Programming and Object Based Programming. Explain some of the major features of C# language. [2+3]
3. Explain overview of Microsoft .NET framework and its components in detail. [5]
4. What do you mean by property in C# language? How is it different from method? Compare automatic property with other types of property with suitable example. [1+1+3]
5. Define constructor. Explain different types of constructors used in C# with example. [1+4]
6. Define inheritance. Write a C# program to demonstrate multilevel and multiple inheritance. [1+2+2]
7. What is generics? List different types of generic classes. Explain delegate with example. [1+1+3]
8. What do you mean by lambda expression? Explain different types of lambda expression used in C# with example. [1+4]

Group C

Attempt any TWO questions. [2x10=20]

9. a) Write a program to create user registration form in one ASP.NET web page and display filled data in another page. [7]
b) Write a program for handling exception in ASP.NET. [3]
10. a) How virtual method is used to achieve dynamic binding in C#? Explain with the help of suitable program. [1+4]
b) Define operator overloading. Write a C# program to overload binary operator. [1+4]
11. a) What is LINQ? Write a program to select employees whose salary is greater than 20000 and whose address is Kathmandu using LINQ. [1+4]
b) Write a C# program to show insert and select operation in database. [5]

Course Title: **DotNet Technology (3 Cr.)**

Course Code: **CACS302**

Year/Semester: **III/V**

Class Load: **6 Hrs. / Week (Theory: 3 Hrs., Practical: 3 Hrs.)**

Course Description:

This course covers different concepts of .NET framework. It also covers basic to advanced features of C# language including language basics, creating types and inheritance, delegates, events, lambda expressions, LINQ, working with databases, and developing web applications using ASP.NET.

Course Objectives:

The primary objective of this course is to provide concepts of .NET framework and different concepts of C# programming language and make students familiar with their uses and applications.

Course Contents:

Unit 1: Introducing C# and the .NET Framework (7 Hrs.)

Object Orientation; Type Safety; Memory Management; Platform Support; C# and CLR; CLR and .NET Framework; Other Frameworks; Framework Overview; .NET Standard 2.0; Applied Technologies

Unit 2: The C# Language Basics (12 Hrs.)

Writing Console and GUI Applications; Identifiers and Keywords; Writing Comments; Data Types; Expressions and Operators; Strings and Characters; Arrays; Variables and Parameters; Statements (Declaration, Expression, Selection, Iteration, and Jump Statements); Namespaces

Unit 3: Creating Types in C# (12 Hrs.)

Classes; Constructors and Deconstructors; this Reference; Properties; Indexers; Static Constructors and Classes; Finalizers; Dynamic Binding; Operator Overloading; Inheritance; Abstract Classes and Methods; base Keyword; Overloading; Object Type; Structs; Access Modifiers; Interfaces; Enums; Generics

Unit 4: Advanced C# (14 Hrs.)

Delegates; Events; Lambda Expressions; Exception Handling; Introduction of LINQ; Working with Databases; Writing Web Applications using ASP.NET

Laboratory Work: The laboratory work includes writing console and/or GUI programs in C#

- To implement basic language features
- To create classes and objects and to implement different object-oriented features
- To implement inheritance
- To implement advanced features like delegates, event handling, lambda expressions, exception handling
- To implement LINQ and database applications

2018
Bachelor in Computer Application (BCA)
Course Title: DotNet Technology
Code No: CACS 302
Semester: V

Group B

Attempt any Six questions :

2. What is CLR? List and explain the feature of CLR.
3. What do you mean by value type and reference type? Give an example for each.
4. What is an interface? How interface is used in multiple inheritance? Explain with example,
5. Why namespace is used in C#? How to create and use the namespace in C#? Explain with example.
6. What is an Exception? Write a program to generate divide by zero exception and index out of Range exception and also handle these exception.
7. What is a Query expression? Write a C# program to select Students who are lived in Kirtipur and studied in Patan Multiple Campus using LINQ
8. What do you mean by static class and constructor? Explain with suitable example.

Attempt any Two questions:

9. What do you mean by operator overloading? Write a C# program to demonstrate binary and relational operator overloading.
10. Explain difference between ExecuteReader and ExecuteNonQuery. Provided that a mysql database named "Company" with table named "Product" with following columns (ProductId as int, ProductName as varchar(20), and UnitPrice as float). Write a cl program to connect to the database and display product records that have UnitPrice is greater than RS. 1000 and update the product record (such as UnitPrice=4500) from product database whose ProductId=50.
11. What is web server control in ASP.NET? List five web server control used in ASP.NET. Write a program to create student registration form in one ASP.NET page and display the filled data with another page.

Tribhuwan University
Faculty of Humanities and Social Sciences
OFFICE OF THE DEAN
2020
Bachelor of Computer Applications

Course Title: DotNet Technology
Code No: CACS302
Semester: Fifth Semester

Group B

2. Differentiate Object Oriented Programming and Object Based Programming. Explain some of the major features of C# language. [2+3]
3. Explain overview of Microsoft .NET framework and its components in detail. [5]
4. What do you mean by property in C# language? How it is different from method? Compare automatic property with other types of property with suitable example. [1+1+3]
5. Define constructor. Explain different types of constructors used in C# with example. [1+4]
6. Define inheritance. Write a C# program to demonstrate multilevel and multiple inheritance. [1+2+2]
7. What is generics? List different types of generic classes. Explain delegate with example. [1+1+3]
8. What do you mean by lambda expression? Explain different types of lambda expression used in C# with example. [1+4]

Group C

9.
 - a) Write a program to create user registration form in one ASP.NET web page and display filled data in another page.
 - b) Write a program for handling exception in ASP.NET.
10.
 - a) How virtual method is used to achieve dynamic binding in C#? Explain with the help of suitable program. [1+4]
 - b) Define operator overloading. Write a C# program to overload binary operator.
11.
 - a) What is LINQ? Write a program to select employees whose salary is greater than 20000 and whose address is kathmandu using LINQ.
 - b) Write a C# program to show insert and select operation in database.

Tribhuvan University
Faculty of Humanities and Social Sciences
OFFICE OF THE DEAN
2024
Bachelor of Computer Applications

Course Title: DotNet Technology
Code No: CACS302
Semester: Fifth Semester

Group B

Attempt any SIX questions.

2. What is managed code? Give reason to justify the statement "DOT NET framework is platform independent an Language interoperability".
3. What is optional parameter? Write a C# program which stores values in two enumerations, Department and College. It uses two functions to display the data contained in Department and College enumerations.
4. What is difference between value type and reference type in C#? Explain the concept of boxing and unboxing with suitable program.
5. What is interface? Explain with suitable program, how interface can be used to perform multiple inheritance?
6. What are use of finalizer in C#? Explain with suitable program, how base keyword is used to call member function and constructor of parent class?
7. What are keyword used in exception handling? Write a C# program to handle IndexOutOfRangeException and InvalidCastException.
8. What is query syntax in LINQ? Explain the standard query operators "select", "contains", "orderBy" and "where" in LINQ with suitable example.

Group C

Attempt any TWO questions.

- 9.a) What is an event? Explain with suitable program, how event is handled in C#.
- b) What is Lambda expression? Explain the concept of generic delegates with suitable program.
10. How this keyword is used to declare indexer? Write C# program to overload unary (++) and relational(==) operator.
11. What are component of [ASP.NET](#)? Explain various validation server control in [ASP.NET](#) with suitable program.

Tribhuvan University
Faculty of Humanities and Social Sciences
OFFICE OF THE DEAN
Practical Examination 2024

Bachelor in Computer Application
Course Title: Dot Net Technology
Code No: CACS 302
Semester: V

Full Marks: 20
Pass Marks: 8
Time: 1.5 hrs.

Set-A

Attempts any THREE questions

5. Write a C# console application to define **Person** class and derive **Student** and **Employee** Class from it to demonstrate hierarchical inheritance.
6. Write a C# program to define class **Distance** with attribute **feet and inches** and then add and subtract the two distances using binary operator overloading.
7. Write a C# program to display Employee records in descending order with respect to employee name using LINQ.
8. Write a c# program to connect to the database and insert 5 student records and display the student record from College database whose faculty is "Computer Application". (Database: College Table: Student (Roll, Name, And Faculty))

Tribhuvan University
Faculty of Humanities and Social Sciences
OFFICE OF THE DEAN
Practical Examination 2024

Bachelor in Computer Application
Course Title: Dot Net Technology
Code No: CACS 302
Semester: V

Full Marks: 20
Pass Marks: 8
Time 1.5 hrs.

Set-B

Attempts any THREE questions

1. Write a C# program to define class **Employee** with attribute **Eno, Ename, Address, and Salary**. Set default and user-defined value for these attribute using Constructor and then display values of these attribute.
2. Write a C# program that will read balance and withdraw amount from keyboard and display the remaining balance on screen if balance is greater than withdraw amount otherwise throw an exception with appropriate message.
3. Write a C# program to select Students who are lived in **Kirtipur** and studied in **RR Campus** using LINQ
4. Write a c# program to connect to the database and insert 3 Employee records and delete the Employee record from company database who's Department is "Sales".
(Database: Company Table: Employee (EId, Ename, Salary, Department))